

1(a)(i)	A = endodermal (cell) ; B = xylem vessel element ; A vessel elements A xylem elements I xylem	2
1(a)(ii)	(has cells that can) divide continuously by mitosis ; <i>idea that</i> forms cells that can, differentiate / AW, into cells in, vascular tissue / xylem (tissue) / phloem (tissue) ; (for) repair / growth, of, vascular tissue / xylem / phloem ; divide to maintain pool of procambial cells ;	1

Question	Answer	Marks
1(b)	<i>mark as pairs (adaptation A + explanation E), max two pairs</i> A thick (waxy) cuticle ; E increased / long, diffusion distance for water <u>vapour</u> or <i>idea of</i> (greater) impermeability to, water <u>vapour</u> ; A needle-shaped / narrow / AW, leaves ; E low surface area to volume ratio, qualified ; e.g. less transpiration / AW <i>allow ecf if adaptation is spines</i> A multilayered epidermis / hypodermis ; E increased diffusion distance for water <u>vapour</u> ; A low stomatal density ; A few(er) stomata (per unit area) A small(er) stomata E less, transpiration / diffusion of water vapour out of the plant (because most water loss is via stomata) ; I evaporation A sunken stomata ; A other examples e.g. stomata in, grooves / crypts / chambers A trichomes / (stomatal) hairs; A rolled / curled, leaves ; <i>for above three mps</i> E maintains humid air around stomata / reduced water potential gradient / (creates) still / non-moving, air ; I concentration gradient AVP ;; e.g. A midday closure of stomata E close at times when transpiration is highest / AW A close packing of mesophyll cells / fewer (intercellular) air spaces E reduces evaporation from mesophyll A stomata open at night E lowers transpiration as, higher humidity / lower temperature A lighter gray / pale coloured, leaves E reflect light to keep temperature cooler	4